

AI-Augmented Productivity for Finance

IESE MiF · Prof. Sara Bisbe

- **The goal of the two days:** a working AI financial analyst, built by you
- **Format:** you build it on your own laptop, session by session
- **The result is a portfolio:** everything you build leaves with you

Part 1 · Course overview

Five sessions, one system

Session	You build	The skill
1 · Prompting	grounded prompts	a model that refuses rather than guesses
2 · Coding copilot	Apple valued against its peers	direct Claude Code; verify its work
3 · Debugging & analytics	that valuation, repaired + an earnings engine	prove a number is right
4 · Workflows	a screening workflow on live SEC filings	code decides; the model explains
5 · Agents	a governed agent, published on GitHub + capstone	the model plans, within your limits

Agenda

	Monday	Tuesday
9:00	1 · Prompting and financial reasoning	4 · Workflows on live SEC data
10:30	2 · Claude Code as coding copilot	5 · Agents + your capstone
12:00	3 · Debugging , testing, analytics	

- **Capstone:** 5 minutes per person, Tuesday. Your workflow, running, on screen.

The libraries you will use

	Library	Standing
Data	pandas · numpy	the industry standards for data and numerical work
Charts	plotly	best in class for interactive visuals
Testing & validation	pytest · pydantic	the standard test runner; the standard validation library
Claude	anthropic · PydanticAI	the official SDK; a state-of-the-art agent framework
Workflows	LangGraph	the industry standard for AI pipelines, from the LangChain ecosystem

Each appears exactly when its job appears — nothing is used before it is explained

Part 2 · How every session works

The session rhythm, identical all five times

I explain → we run it → you build → a check confirms it is right
(concepts) (live) (the lab) (green means continue)

Segment	Minutes
Concepts, with runnable demonstrations	10–15
Live demonstration	15–22
Your lab — the largest block	25–30
Debrief and bridge to the next session	5–8

How the labs work

Every exercise sits between two markers. **You fill the gaps. Nothing else changes.**

```
### START CODE HERE ###  
df["revenue_growth_1y"] = df[None] / df[None] - 1  
### END CODE HERE ###
```

- **None** → replace with the correct column, value or variable
- **[QUESTION IN CAPITALS]** → replace with the text the bracket asks for
- Run the  **check cell** below each exercise; green means correct, continue
- **When you are stuck:** ask the **✳ Claude panel** — using it well is part of what this course teaches

Asking Claude about specific code

Option+K (Mac) / **Alt+K** (Windows) — the one shortcut to learn today.

1. highlight the lines you are asking about
2. press Option+K → the selection lands in the * panel with its address: file, cell, exact line numbers
3. type your question next to it, press Enter

- The answer is about **exactly those lines**, not about code in general
- Ask for the **cause before the fix**: *"explain why this fails before proposing a change"* — you learn the diagnosis, not just the patch
- Claude's edits arrive as a **diff**: read it, then accept or reject
- Shortcut not working? Copy the lines into the panel by hand — same result

Your two AI tools, and how to tell them apart

	✱ Claude panel	API key in <code>.env</code>
What	Your copilot in VS Code	Claude answering your code
Paid by	Pro subscription	Your \$5 credit
Used for	Help while building	The tools you build

API stands for application programming interface: the channel through which your own code calls Claude. If a chat panel mentions GPT, Gemini or credits, it is VS Code's built-in assistant, not the course tool. The course tool is always the ✱ panel.

Part 3 · Setup — first 15 minutes, in class

Setup: your first commands

Open VS Code → **Terminal** → **New Terminal**, and run:

```
git clone https://github.com/comtessa-sareta/iese-ai-finance-bootcamp-student.git  
cd iese-ai-finance-bootcamp-student
```

Then: **File** → **Open Folder** → choose `iese-ai-finance-bootcamp-student`, open `setup/SETUP.md`, and follow it top to bottom — it contains every remaining command, in order.

- **Both lines matter:** `clone` downloads the folder, `cd` moves you *into* it. Every later command runs from inside — `ls` should show `requirements.txt`
- The link is also in the **Zoom chat**: click it there instead of typing
- `git` not found? `SETUP.md` step 0b covers installing it

What SETUP.md walks you through

Step	What	You already have
0	Accounts: Claude Pro + Anthropic API key (\$5 credit)	Pro ✓ — a lost key is re-created in 30 seconds
1	Clone the course repository	Git, from the tools pre-course
2	Create the <code>aifinance</code> Python environment	Anaconda, from the pre-course
3	Put your keys in <code>.env</code>	—
4	Sign in to the ✱ Claude panel in VS Code	—
5–6	Verify: checker all <code>[OK]</code> , then notebook <code>00</code> all green	—

Setup: the finish line, and helping each other

- **Finished and green?** Raise your hand — then help a neighbor
- **Stuck more than 3 minutes?** Work on your neighbor's laptop; we repair yours at the break — nobody sits out
- Every failure message in the checker includes its own fix; the Troubleshooting section of `SETUP.md` covers the rest

The one rule of this course

Never ship a number you haven't verified.

- In finance, a confidently delivered wrong number carries serious professional consequences
- Everything we build makes verification **cheap, fast, automatic**